GreenThreads : An AI-Powered Sustainable Fashion Platform for Recycled Clothing

Paper based on project

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Abstract — **The fashion industry is a significant contributor to environmental degradation due to the effects of fast fashion. In this paper, an AI-powered sustainable e-commerce system named GreenThreads is proposed, which encourages the buying and controlled selling of 100% recycled fashion products. The system utilizes the intelligent capabilities of Machine Learning (ML) microservices with Node.js, providing intelligent features to the system. A novel sustainability measure is introduced, quantifying the environmental benefit achieved with each purchase. Moreover, controlled selling is achieved by only accepting bulk clothing contributions that satisfy certain conditions, ensuring efficiency.**

Keywords — **Sustainable Fashion, Machine Learning, FastAPI, Recycling, Sentiment Analysis, Recommendation System, Customer Segmentation**

I. INTRODUCTION

The fashion industry is a significant contributor to environmental pollution and textile waste due to its linear production pattern of use and disposal. Therefore, sustainable practices such as recycling, reusing, and reselling have become significant solutions for mitigating such problems. The paper aims to introduce a new platform named GreenThreads that facilitates sustainable fashion by allowing customers to purchase recycled products and contribute second-hand clothing in bulk, following a set of conditions. The platform provides a sustainable lifecycle for the products and helps in reducing waste and promoting sustainable fashion consumption patterns.

However, despite increasing awareness about sustainable fashion practices, the phenomenon remains low due to several concerns such as trust, hygiene, and quality. The proposed platform will address these concerns by implementing AI-based quality checks and admin verification for ensuring high-quality and authentic products. The platform will also be engaging and informative due to the inclusion of sustainability scores.

The system is designed based on a modular structure with modules related to user management, product management, order management, and machine learning services. The main goal is to encourage sustainable fashion, facilitate controlled selling by applying bulk conditions, guarantee product quality, improve user experience, and enable machine learning-based decision-making. Moreover, the system is designed based on features related to scalability, data-driven insights, and business viability in terms of pricing and user engagement strategies.

II. PROBLEM STATEMENT

The rapid growth of the fashion industry, especially fast fashion, results in a significant increase in textile wastes, pollution, and unsustainable consumption patterns. Although there are a number of platforms that offer the facilities of buying and selling second-hand fashion products, these platforms are not equipped with proper quality control, sustainability, and smart decision-making tools. In addition, the selling option may result in inefficiency, quality, and logistical problems.

There is a need for a smart and sustainable system that not only supports the use of recycled fashion products but also focuses on quality, user participation, and the utilization of various technologies to improve user experience and efficiency. The existing platforms are not capable of effectively combining sustainability practices and smart machine learning capabilities.

III. LITERATURE SURVEY AND EXISTING SYSTEMS

Sustainable fashion has received significant attention in recent times due to the environmental and social implications of traditional fashion business models. The application of Artificial Intelligence in sustainable fashion has received significant attention in recent research studies, which focused on areas such as reduction of wastes, supply chain management, analysis of consumer behavior, and sustainable design.

From the analysis of these studies, it is clear that AI is being effectively used for demand prediction, resource optimization, eco-material recommendation, trend forecasting, and customer engagement. However, most of the existing studies are conceptual frameworks and literature surveys, and very little empirical work is done in the context of sustainable fashion practices.

Table 1 shows summary of seven significant research studies on the application of Artificial Intelligence in sustainable fashion, along with their focus areas, contributions, shortcomings, and sources.

TABLE 1: LITERATURE REVIEW

Reference / Paper Name	Limitations	How Our Recycled Fashion Application Overcomes It	Content Summary / Focus
A Systematic Review of AI for Circular Fashion (2025)	Small datasets; lacks whole-garment integration	Integrates full garment tracking with ML for better recycling efficiency and material reuse	AI in circular economy & textile recycling; sorting, quality tracking
AI & Sustainability in Fashion (2010–2022)	High data costs, implementation complexity	Uses lightweight ML models and open datasets for efficient deployment in recycled fashion production	Supply chain optimization, waste control, trend forecasting
AI for Sustainable Fashion Design & Supply Chains (2026)	Limited real-world evaluation	Implements real-time production monitoring and AI-assisted sustainable design suggestions	AI frameworks for design optimization and supply efficiency
Predictive Waste Reduction & Eco-Material Optimization (2025)	Limited empirical results	Our app tracks material usage and suggests recycled or upcycled fabrics to minimize waste	Reducing fabric waste and recommending eco-friendly materials
AI-Based Conversational Agents for Sustainable Fashion (2025)	Only focuses on conversation, not production	Integrates consumer recommendations with backend supply chain data to ensure sustainable production and purchase decisions	Chatbots guiding sustainable consumer choices
AI-Driven Consumer Research Review (2026)	Focuses on consumer side, little on production	Connects AI-driven recommendations with recycled material availability, creating a full-cycle sustainable solution	AI personalization and consumer engagement in fashion e-commerce
AI Forecasting Sustainable Fashion Trends (2025)	Limited experimental validation	Our system validates trend predictions with actual recycled material availability and user engagement	ML/NLP to predict eco-fashion trend adoption

The current approaches in the implementation of AI in sustainable fashion have mainly focused on individual aspects of the fashion supply chain, not on the overall system.

- Circular Fashion & Recycling: The current system has employed AI for textile sorting and quality control. However, there is no overall garment integration.
- Supply Chain Optimization: The current AI framework has optimized production schedules to minimize overproduction. However, testing these systems in the real world is challenging.
- Consumer Awareness & Engagement: The current system has employed chatbots and personalization techniques for consumer awareness of sustainable fashion. However, these are limited to front-end implementation only.
- Design Innovation: The current system has employed generative AI for sustainable design and trend forecasting. However, there are challenges in the availability of data for these techniques.

Overall, the current system has focused on individual aspects of sustainable fashion, such as design, supply chain, or consumer awareness. However, there is no integrated framework for the implementation of sustainable fashion. Thus, there is a need for the development of such a system, which has led to the proposed system in this research.

IV. METHODOLOGY AND IMPLEMENTATION

The GreenThreads platform is developed through a structured approach that brings together sustainability and state-of-the-art machine learning techniques. The overall aim is to develop a system that allows for efficient reuse of waste textiles, along with a seamless and intelligent experience for users. The first step in this approach is for users to interact with the system. This is done through users purchasing recycled products or contributing used clothes in bulk. When clothes are contributed for selling, users go through a system that checks for parameters like quantity and quality. This ensures that only suitable clothes are sent forward for recycling and production. Machine Learning plays an important part in enhancing different parts of the system. For instance, image-based machine learning is used for detecting fabric and defects, thus maintaining the quality of products. Another part is customer segmentation, which helps in grouping users according to their purchasing behavior. This enhances the overall experience for users. The system also allows for image similarity search, through which users can search for products similar to the image uploaded. There are also other features like demand forecasting and dynamic pricing, which will be implemented in the future and will enhance decision-making for users. The development of the GreenThreads system is through a modular and scalable approach. The system is divided into three parts: frontend, backend, and Machine Learning (ML).

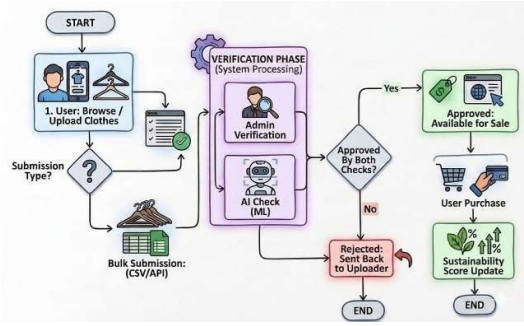

Fig.1 Workflow for item listing and verification

The frontend is designed to offer a simple user interface through which users can interact with the system by viewing products, adding clothes, and viewing sustainability-related information. The backend, which is developed using Node.js, is responsible for handling key operations such as user authentication, product handling, order handling, and interaction with the ML service.

The ML service is designed as a separate entity using FastAPI. It contains various modules for handling various features such as classification, visual search, and segmentation. Each of the features is independent, making the system easier to manage, maintain, and improve. These features can be accessed through APIs, making it easier to communicate with the backend.

Each of the ML features is designed to have separate folders with separate files for handling operations such as training, testing, and prediction. For instance, ResNet can be used for image classification, whereas YOLO can be used for defect detection.

All key operations such as user handling, product handling, order handling, etc., are maintained in a centralized database. Scripts are also used for operations such as data handling, periodic updates, etc., such as customer segmentation.

The implementation of all the components in a well-organized manner, with each component separate from the others, ensures that the system is flexible, efficient, and ready for improvements.

The overall design, methodology, and implementation of GreenThreads are aimed at ensuring that sustainability is combined with smart technologies to offer a practical solution to the challenges in the fashion industry.

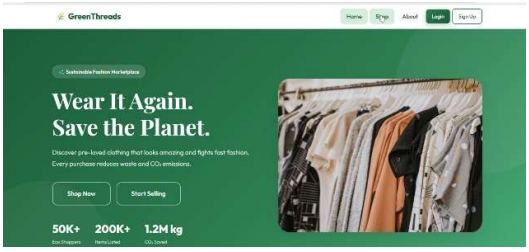

Fig.2 Landing Page

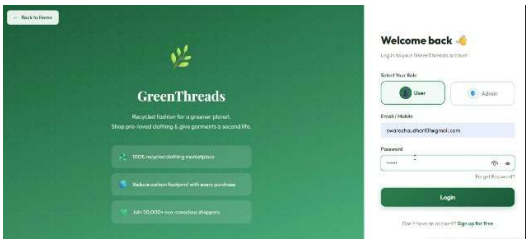

Fig.3 Login Page

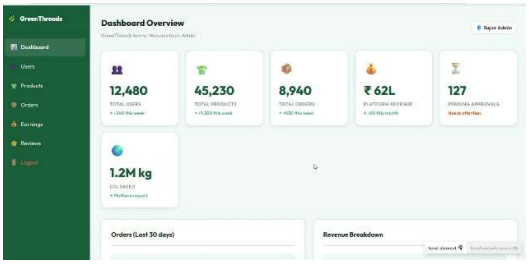

Fig.4 Admin Dashboard

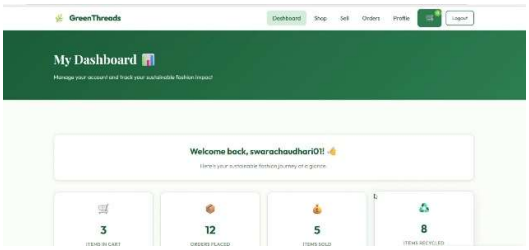

Fig.5 User Dashboard

V. SYSTEM ARCHITECTURE

The GreenThreads platform follows a **microservices-based architecture** designed to ensure scalability, flexibility, and efficient performance. The system is divided into three main layers: **Frontend, Backend, and Machine Learning (ML) Service**, each responsible for specific functionalities.

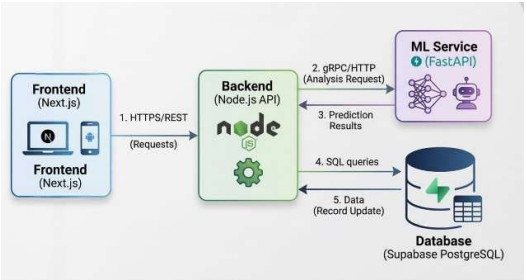

Fig.6 System Architecture

1. Frontend Layer

The frontend acts as the user interface of the platform. It allows users to:

- Browse recycled products
- Upload clothing for selling
- View sustainability scores and product details
- Interact with recommendations and search features

It provides a simple and user-friendly experience while sending user requests to the backend.

2. Backend Layer (Node.js)

The backend is responsible for handling all core operations of the platform. It acts as a bridge between the frontend and the ML service.

Main responsibilities:

- User authentication and authorization
- Product and inventory management
- Order processing and tracking
- Managing bulk selling requests and admin approval
- Communicating with the ML service via HTTP APIs
- Storing and retrieving data from the database

The backend ensures smooth coordination between different components of the system.

3. Machine Learning Service (FastAPI)

The ML service is a standalone application built using FastAPI and runs independently on a separate server (port 8000). It provides intelligent features through API endpoints.

Key modules include:

- **Textile Classification:** Identifies material type and assigns quality grade
- **Visual Search:** Finds similar products using image embeddings
- **Customer Segmentation:** Groups users based on purchasing behavior
- Demand Forecasting, Recommendation System, Dynamic Pricing
- Defect Detection for quality inspection

Each feature is implemented in a separate module, allowing independent development and easy maintenance.

4. Database Layer

A centralized database is used to store:

- User information
- Product details
- Order history
- Selling requests and approval status
- ML outputs such as classifications and segments

This ensures efficient data management and smooth tracking of the product lifecycle.

5. Communication Flow

The interaction between components follows this flow:

User → Frontend → Backend → ML Service → Backend → Database → Frontend

- The frontend sends requests to the backend
- The backend processes the request and calls the ML service if needed
- The ML service returns predictions or results
- The backend stores results in the database and sends the response back to the frontend

6. Key Advantages of Architecture

- **Scalability:** Each component can be scaled independently
- **Modularity:** Features are isolated and easy to maintain
- **Flexibility:** New ML features can be added without affecting the system
- **Performance:** Efficient handling of user requests and ML computations

Overall, the system architecture of GreenThreads is designed to provide a seamless user experience while efficiently integrating intelligent features and ensuring long-term scalability.

VI. TECHNOLOGY STACK

The GreenThreads platform is built using modern technologies to ensure scalability, smooth user experience, and intelligent AI capabilities. The stack is organized by system layers:

1. Frontend

- **Technology:** Next.js
- **Purpose:**
  - Provide a fast, interactive, and SEO-friendly interface
  - Display products, sustainability scores, and user dashboards
  - Allow image uploads, browsing, and seamless navigation

2. Backend

- **Technology:** Node.js
- **Purpose:**
  - Handle user authentication and authorization
  - Manage products, orders, and selling requests
  - Communicate with ML services via HTTP APIs
  - Implement admin verification workflows

3. Machine Learning Service

- **Technology:** FastAPI + Uvicorn
- **Purpose:**
  - Serve AI models through REST APIs
  - Handle textile classification, defect detection, visual search, and segmentation

4. Machine Learning & AI Libraries

Layer	Library	Version	Purpose
Deep Learning	PyTorch + Torchvision	2.3.0	Image-based feature extraction & model training
Computer Vision	ResNet-50, YOLOv8	–	Textile classification & defect detection
Classical ML	Scikit-learn	1.5.0	Customer segmentation & preprocessing
Gradient Boosting	XGBoost	2.0.3	Dynamic pricing and predictive analytics
Vector Search	FAISS (CPU)	1.8.0	Visual similarity search
Time Series	Prophet	1.1.5	Demand forecasting
Explainability	SHAP	0.45.0	Model interpretability

5. Data Processing

- **Libraries:** Pandas (2.2.2), NumPy (1.26.4)
- **Purpose:** Cleaning, transformation, and analysis of structured and unstructured data

6. Image Processing

- **Library:** Pillow (10.3.0)
- **Purpose:** Image handling, preprocessing, and resizing

7. Database

- **Technology:** Supabase (PostgreSQL) + Prisma ORM
- **Purpose:**
  - Store users, products, orders, and ML outputs
  - Enable type-safe and efficient database operations via Prisma
  - Track recycling and sustainability metrics

8. Communication

- **Protocol:** HTTP / REST APIs
- **Library:** httpx (latest)
- **Purpose:** Enable seamless communication between backend and ML service

9. Development & Utilities

- **Version Control:** Git & GitHub
- **Environment Management:** .env configuration
- **Automation Scripts:** Python scripts for training, evaluation, and periodic updates

This stack ensures GreenThreads is **fast, modular, scalable**, and **capable** of integrating advanced AI features while maintaining a smooth and engaging user experience.

VII. TESTING AND VERIFICATION

To ensure that GreenThreads functions reliably, efficiently, and securely, a comprehensive testing and verification strategy is implemented. Testing covers every component of the platform: the frontend, backend, ML service, database, and their integrations.

1. API Testing

All REST APIs, including those for product management, user authentication, order processing, and ML predictions, are thoroughly tested using **Postman**. The testing process ensures that:

- Each endpoint returns correct responses under normal and edge cases.

- Error handling works correctly, providing meaningful messages for invalid inputs or unauthorized actions.
- Integration between the backend and ML service is seamless, and ML predictions are returned accurately.
- The controlled selling workflow is enforced properly—users cannot bypass quantity checks or admin approvals.

Postman collections are maintained for **regression testing**, ensuring that new updates do not break existing functionality. Automated scripts can also be integrated to run these tests periodically.

2. Frontend Testing

The **Next.js frontend** is tested to guarantee a smooth and responsive user experience:

- Navigation across pages is seamless and intuitive.
- Product uploads, searches, and purchases work without glitches.
- Sustainability metrics, eco-scores, and personalized recommendations display correctly.
- Form validations prevent incorrect user inputs (e.g., missing product details or invalid quantities).

3. Backend Verification

The **Node.js backend** undergoes rigorous verification to ensure reliability and data integrity:

- Authentication and authorization checks protect sensitive user and admin data.
- Product and order management workflows are verified for correctness.
- Communication with the ML service is verified through endpoint testing, response validation, and latency checks.
- Controlled selling logic is tested to prevent bypassing bulk quantity or quality checks.

4. Machine Learning Service Testing

The FastAPI-based ML service is tested independently to ensure accurate predictions:

- **Textile Classification:** Image inputs are verified to ensure the correct fabric type is detected.
- **Defect Detection:** YOLOv8 outputs are validated against a dataset of faulty and flawless products to ensure accuracy.
- **Visual Similarity Search:** Similar items returned are cross-checked for relevance.
- **Customer Segmentation:** User groupings are tested against sample datasets to ensure meaningful segmentation.

Unit tests and sample datasets are used to validate models before deployment.

5. Database Verification

The PostgreSQL database (via Supabase) is tested for:

- Correct storage of user data, product details, orders, and ML outputs.
- Consistent tracking of product lifecycle from purchase to recycling.
- Integrity and consistency during concurrent operations (e.g., multiple users uploading items simultaneously).

6. End-to-End Testing

The platform is also verified end-to-end:

- Users can successfully browse products, purchase items, and contribute clothing.
- Admin approvals and controlled selling workflows function as intended.
- Sustainability scores, credits, and recommendations are correctly calculated and displayed.
- Overall system performance, responsiveness, and stability are monitored under realistic usage scenarios.

7. Continuous Verification

- Regression testing is performed whenever new features or updates are deployed.
- Logs and monitoring tools track API performance, errors, and unusual patterns for proactive debugging.
- Feedback from pilot users is incorporated to improve system usability and trustworthiness.

VIII. CONCLUSION

Thus, the GreenThreads platform is an example of an effective and efficient way of promoting green fashion and sustainability. The platform's integration of e-commerce capabilities with recycling, resale, and return options enables the creation of a circular and closed-loop system that maximizes the product life of clothing items and minimizes waste.

Moreover, the platform's capabilities, such as eco-score, credit rewards, and lifecycle tracking, motivate and engage users in green practices and activities. However, the platform also presents an effective business model that balances sustainability with business profitability through low-cost business models.

Although the platform faces challenges in terms of user participation and business costs, it presents a solid foundation for future growth and success. The platform's integration of other cutting-edge technologies, such as AI and predictive analytics, could further improve its effectiveness and efficiency in creating a smarter, greener, and more sustainable fashion ecosystem.

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